

Kaun Lab Protocol

Protocol Name: Kaun Lab_ Two-Choice assay_TREX_+_data aggregation code

Date: 1/26/2026

Version:v.2

Protocol prepared by: Ari Bouchie and Sam Stea

Adapted from: Version v.1

Purpose of this protocol: This protocol will help with analyzing the videos from FlyGrAMs 2 Choice. If any issues occur when using Ctrex or data aggregation code, contact Tariq Brown or Sam Stea.

Ctrex

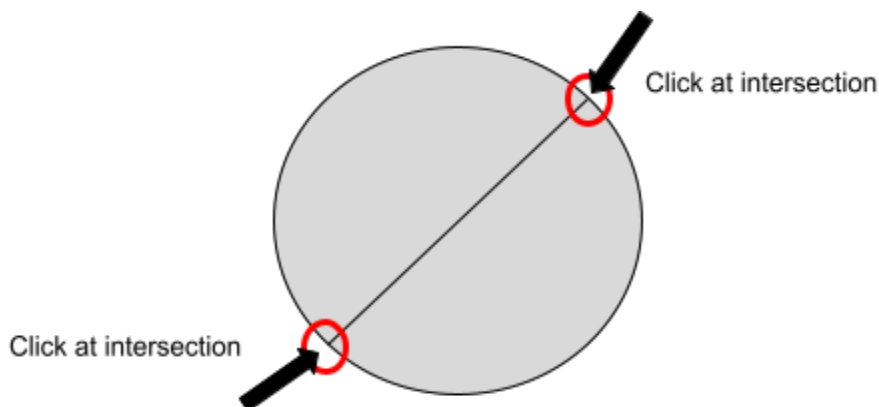
Setting up for running Trex

1. On computer with Trex on it, transfer folder with all Fly gram videos
2. Make two folders- 'FlyGramVideos' and 'TrexOutputs'
 - a. In 'TrexOutputs' make folders for all the timestamps that are from the FlyGramVideos for example (11.43.28)
 - b. In the time stamp videos make a 'Bottom' and 'Top' folder (case sensitive)
3. Open anaconda on computer
4. Once that loads, click and open powershell prompt
 - a. If a window of anaconda was already open, it won't work. Close the window that was open and reopen a new one
5. Click '↑' twice
6. Click 'Enter' twice
7. '↑' twice again
8. 'Enter' once

To process bottom arena of video

1. Convert raw
2. Click '..' at top of window
3. 'Desktop'
4. Find your folder
5. Click into folders and find the video you want to process (Video in FlygramVideos)
6. Change these settings ONLY
 - a. Average_samples → 250
 - b. Averaging_method → mode
 - c. Threshold → 25
7. Click 'open' and once you start to hear the fan in the computer that means the video is done and you can proceed
8. '↑' once

9. 'Enter'
10. In the white box at the top of the screen, look up the name of your video by typing in "video— Specific date" and click the correct timestamp
 - a. The date you type in is the date on the video you are running
11. Change this setting ONLY
 - a. Track_max → 10
12. Click 'Open'
13. In left white box type 'output' and click 'output_format'
14. Click 'csv'
15. Go back to where you typed output and click on output_dir
 - a. Go to 'TrexOutputs' folder on computer and click on corresponding timestamp folder with the FlyGram video
 - b. Click into 'Bottom' folder
 - c. 'Copy address'
16. Replace the automatic directory Trex puts in by pasting the address into the white box to the right of where you typed output
17. 'Enter'
18. Go back to output box and now click 'output_csv_decimals'
19. Change adjacent right box to '10'
20. 'Enter'
21. Hold down 'ctrl' and click four spots around the bottom right arena on the video so it makes a box
 - a. Don't get any of the top arena
22. Click 'append to...'
23. 'Track_include'
24. Click 'reanalyze' at top of window
25. Wait until it processes, and then click ctrl-S to save
26. Now you need to set reference, so hold ctrl while you make a line across the bottom left arena

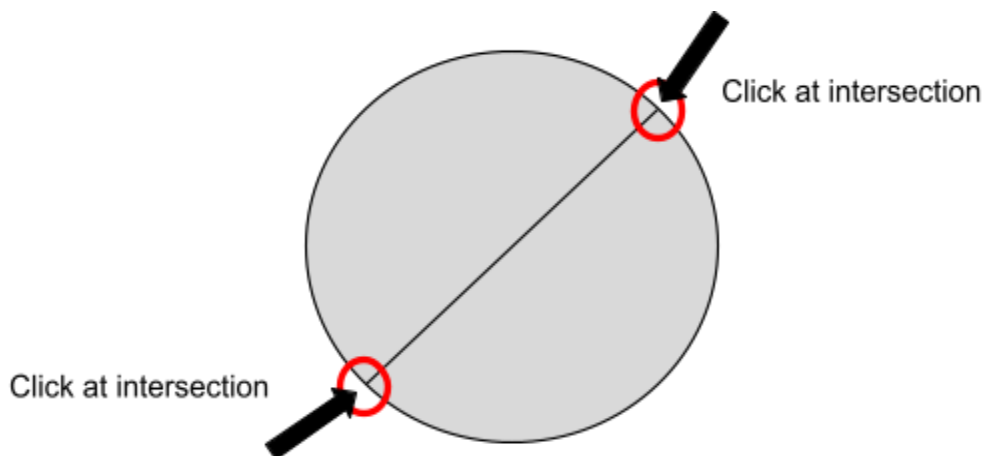


27. Click 'use known length to calibrate'
28. Type in 3.6

29. 'Okay' and then 'okay' again
30. Ctrl-S to save progress
31. Go into the timestamp 'Bottom' folder you saved to and check to make sure all the files saved as the same size
 - a. Most of the time one of the 10 files will save as 1kb which we don't want. If you see this go back to the screen where you see the arenas and 'scrub' the video by pressing '→' through the whole video. Once it gets the end of the video click '←' until it gets back to the beginning of the video
 - b. Ctrl-S again and make sure all the files are now the same size
32. Click into one of the files and it will take you to an excel sheet
33. Check the 'Q' column to see if the numbers are lower than 10
 - a. If not make a note of it. Values significantly greater than 10 mean that the two-choice was not placed directly below the arena during that trial, and may need to be excluded.

To process top arena of video

1. Where it says 'output_csv_decimals' delete that and type in track_include
2. Adjacent right box delete everything and type in '[' (no space, just included so you can see both symbols)
3. 'Enter'
4. Type 'output_dir'
5. Hold down 'Ctrl' and click 4 spots around top arena to make a box 'append to...' and click 'track_include'
6. Go to where the directory is and delete just the 'Bottom' in the directory and type 'Top'
7. 'Enter'
8. Hold down Ctrl and make another line to set the reference



9. 'Use known length to calibrate'
10. Type in '10' this time
11. 'Okay' and then 'okay' again
12. 'Reanalyze'
13. Ctrl-S to save
14. Make another line for another reference
15. 'Use known length to calibrate'

16. Put back to '3.6'
17. Ctrl-S
18. Check file sizes that were saved → if 1kb scrub the video
19. Check 'Q' column again
20. To do another video, exit out of the screen that has the arenas in it only
21. '↑' once
22. 'Enter' once
23. Restart with another video and do it until all videos are processed so they can be ran through Tariq's code

Data aggregation code (A.K.A Tariq's Code)

Setting up to run Tariq's code

1. Have a folder set up in your google drive named something that corresponds to what was ran
 - a. Ex: ShockMemoryCS
2. In that folder make a new folder named 'TrexOutputs'
3. In 'TrexOutputs' folder make a folder with the date in this format: 2025-07-16
4. Put all timestamps in 'TrexOutputs' folder into the correct date folder you made in your google drive

Running Code

1. Go to "Data aggregation [code](#)" on github, it will be the latest version. Open/copy it into google colab.
2. Scroll down to 'Setup' and click the play button next to '5 cells hidden'
 - a. Agree to all the pop ups, run anyway, continue with your account
3. Go down to 'Function' and click the drop down arrow
 - a. First subarrow 'cleaning the input csv and designating sides as A or B'
4. We will be analyzing the bottom arena first

```

"""
choose label point 1 is your top well has odor A at the top and b at the bottom, and you bottom well has A at the bottom and b at the top
choose 2 if its the opposite
"""
def label_point(row):
    point = np.array([row['X#wcentroid (cm)'], -row['Y#wcentroid (cm)'])
    # Compute the determinant to check the orientation
    if paradigm == assays[0]:
        det = np.linalg.det(np.array([diagonal_point1 - point, diagonal_point2 - point]))
        if det > 0:
            return 'B' # Point is above the line
        else:
            return 'A' # Point is below the line
    """Apparently doesnt work :D """
    # elif paradigm == assays[1]:
    #
    # det = np.linalg.det(np.array([diagonal_point1 - point, diagonal_point2 - point]))
    # if det > 0:
    #     return 'B' # Point is above the line
    # else:
    #     return 'A' # Point is below the line
    """END """

```

- a.
- b. Make sure first set of A and B is not commented (not all just green) and make sure 'B' #Point is above the line and 'A' #Point is below the line when running bottom arena and make sure both parts of the code are the same

- c. When running the top arena A is above the line and B is below the line and only change the first section of letters (leave the bottom, commented one alone)
- d. Second set of this should stay in green
5. Go back up to 'Function' and relick the drop down arrow so all the sub areas go away and click the play button next to '24 cells hidden'
6. Go to 'Batch Processing' → click drop down arrow → drop down arrow next to '20240925 Batch Function' → Scroll down majority of the way in that box until you find

```
#PLACE YOUR FILES HERE
#1. place the filepath to the folder in between the apostrophes to the right of the "="
directory = '/content/drive/MyDrive/AriBouchieKaunLabDocuments/2ChoiceTrexOutputs/W11182Choice.AmeliasVideos/okfine'
#2. place date in between the apostrophes to the right of the "=" in YYYY-MM-DD format
date = '2025-03-08'
#3. press play to run the box
```

- a. When you get here go to folder icon on the left side of the screen → 'drive' → 'MyDrive' → Go to folder you want to analyze first and get to the 'okfine' folder → click on the three dots next to 'okfine' and then click 'copy path' → paste the directory into where it says directory in the code
 - i. Keep the " around the directory
- b. At very bottom of that box you're in:

```
# diagonal_point1, diagonal_point2, origin_location, port_a_location, port_b_location = arena_designator('Top')
# process_position_folder(directory, date, 'Top')

diagonal_point1, diagonal_point2, origin_location, port_a_location, port_b_location = arena_designator('Bottom')
process_position_folder(directory, date, 'Bottom')
```

- c. Make sure the bottom one is white first
- d. If the bottom one is green like the top one, highlight, press 'ctrl-/'
 - i. When the text is all green it will get overlooked by the code, only stuff in white will be read by the code
 - ii. Want the 'Bottom' to be read first
- e. Press the play button at the top of the box in the code you're in
 - i. This will take a while, but at the end it will say theres an error (an error 10-15 min after pressing play is normal but you don't want an error right away')
- f. Go back to the Functions cell and switch the 'A' and 'B' so A is above the line and B is below the line– only switch the letters that are orange colored
 - i. Click the down arrow again and press play on the '24 cells hidden'

```
# diagonal_point1, diagonal_point2, origin_location, port_a_location, port_b_location = arena_designator('Top')
# process_position_folder(directory, date, 'Top')

diagonal_point1, diagonal_point2, origin_location, port_a_location, port_b_location = arena_designator('Bottom')
process_position_folder(directory, date, 'Bottom')
```

- g. Once that finishes, highlight the bottom part, ctrl/ → now it should be green (opposite to what is in the picture)
- h. Highlight top part, ctrl/ , now it should be white, and press the play button again

7. Go to 'Post-Processing' in the code → drop down arrow → 'path-grabber (separate out based on timestamp!)' *You can only compare two groups of timestamps at a time, it is by default set up to compare preference from a group with the CS+ coming from side A and a group where a CS+ is coming from side B. If you have more than two groups (for example males, females with two separate CS+ groups) you may have to run the Post-Processing more than once in separate directory locations to get specific data combinations you want (EX. if you want to compare female preference for CS#1 coming out of side A to male preference for CS#1 coming out of side A, you could first put all female CS#1 timestamps in side A and male CS#1 timestamps in side B and run it through post processing, then in google drive move these timestamps to a new directory location and in google colab put CS#1 timestamps in side A and female CS#1 timestamps in side B and rerun the post-processing box) and then you would have the male and female preference for side A CS#1 in these two locations to take for further analysis either in a stats software like PRISM or for graphing in the "graphing+choice behavior code."* Be sure to carefully note what group of timestamps are being run and which side preference is measured relative to.
8. Input the directory for post-processing by going to the folder icon again
 - a. Main folder → okfine → date → date folder that was made in the date folder you made earlier → click three dots next to the date folder that was made from running the code → copy path → paste directory
9. Press play in the path grabber box you are in
10. Go to 'generate the odor directories' → paste same directory you put in earlier → press play for that box
11. Go down and click play next to 'this will be the box for the generating mega dataframes and plots associated with them'
12. Go down and click play next to 'for heatmaps and mean+sem line plots'
13. Go down and click play next to 'for region count plots'

You will need a copy of 'graphing + choice behavior' code for additional post-choice behavioral data metrics (like U-turns) and graphing metrics over time like distance travelled per arena side, bout duration, and the combined preference index over time (PIOT). Code and instructions on <https://github.com/kaun-lab/Two-Choice-Assay>.